Maxime Ferrera

Research Scientist at Ifremer, Underwater Robotics Lab

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Google Scholar

Experience

Ifremer, Underwater Robotics Lab, Research Scientist — La Seyne-sur-Mer, France
- Computer vision and robot perception for underwater robotics.

Oct 2020 – present
6 years 1 month

ONERA, DTIS, Postdoctoral Research Scientist — Palaiseau, France
- Visual-inertial odometry for pedestrian localization in complex environments (ANR MALIN).
- Development of OV^{2}SLAM, a fully online and versatile visual SLAM.

Jan 2020 – Sept 2020
9 months

ONERA, DTIS, Research Intern (6 months) — Palaiseau, France
- Multi-modal 3D modeling for autonomous drones and robots.
- Fusion of an infrared camera with a stereo camera to build 3D thermal maps.
- Visual odometry in visually degraded conditions.

2016 – 2016
1 year

Safran Morpho, Assistant Engineer Intern (3 months) — France
- Integration of fingerprint authentication within the Windows Biometric Framework.

2015 – 2015
1 year

Education

PhD **ONERA DTIS and LIRMM, Université de Montpellier**, Computer Vision and Robotics — France
- Thesis: Monocular Visual-Inertial-Pressure Fusion for Underwater Localization and 3D Mapping.
- Advised by Prof. Vincent Creuze, Dr. Julien Moras and Dr. Pauline Trouvé-Peloux.

2016 – 2019

MEng **Polytech Annecy-Chambéry**, Computer Science and Electrical Engineering — France
- Exchange semester at San Diego State University, California (2015).

2013 – 2016

CPGE **Polytech Orléans**, Preparatory classes — France

2010 – 2013

Awards

ONERA best PhD thesis award — Apr 2019
ONERA
www.onera.fr/fr/rejoindre-onera/prix-des-doctorants

3rd place, CVPR 2020 TartanAir visual SLAM challenge — Aug 2020
CVPR 2020 Workshop on Visual Localization and SLAM
sites.google.com/view/vislocslamcvpr2020/slam-challenge

Teaching

ESIEA, Lecturer, Robotics Perception (5th year) — Ivry-sur-Seine, France
2018 – 2018

École des Ponts ParisTech, Teaching assistant, C++ Programming (1st year) — Marne-la-Vallée, France, 2017 – 2018

ENSTA ParisTech, Teaching assistant, State Estimation (Mastère ILEMS) — Paris-Saclay, France, 2017 – 2018

Institut d'Optique Graduate School, Project supervisor, PIMS student projects (2nd year) — Paris-Saclay, France, 2017 – 2018